🚀 Project IQ Chatbot: Comprehensive Capabilities Overview

Intelligent IT Support Automation Platform *Powered by Advanced AI, React, PostgreSQL, and Project-IQ ITSM*

🏗️ 1. Architecture & Technology Stack

The application is built on a modern, decoupled architecture designed for performance, reliability, and security.

🧠 AI & Backend Layer

- **Conversational Engine:** Advanced AI Engine (NLU + Dialogue Management)
- **Custom Actions Server:** Custom Action SDK (Python)
- **Admin API:** Flask REST API
- **Machine Learning:** Sentence-Transformers for Semantic Search
- **Database:** PostgreSQL (Stores conversations, analytics, configuration)

🎨 Frontend Layer

- **Admin Dashboard:** React 19 + Tailwind CSS
- **Chat Widget (Embeddable):** React, Framer-Motion, compiled via Vite
- **Icons & UI Elements:** Lucide-React

🔗 Integrations & DevOps

- **ITSM Integration:** Project-IQ ITSM REST API
- **Secrets Management:** Infisical Vault
- **Infrastructure:** Docker, Docker Compose, Nginx Reverse Proxy

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🤖 2. Core AI & Chatbot Capabilities

The core chatbot engine automates IT support natively without requiring human intervention for routine requests.

- **Smart Intent Recognition:** Uses advanced ML classification to accurately understand over 30+ unique IT support intents.
- **Smart Form Inference:** The bot dynamically extracts entities (e.g., hardware type, software name, locations) directly from user text.
- **Single-Message Ticket Creation:**
  - *User says: "Urgent, printer jammed on floor 3"*
  - *Bot immediately recognizes Category (Printer), Priority (High/Urgent), and Location.*
- **Contextual Memory:** Multi-turn dialog handling (e.g., asking for missing information before submitting a ticket).
- **Multi-Lingual Ready:** Designed with a flexible structure capable of integrating German and French localized models.

✨ 3. Frontend Chat Widget (User Experience)

The embeddable frontend chat widget provides a premium user experience with modern UI/UX design.

- **Glassmorphism Aesthetics:** Clean, modern, semi-transparent design with smooth `framer-motion` animations.

- **Dynamic Markdown Support:** Renders rich text, tables, and bold formatting natively within chat bubbles.
- **File Uploads:** Drag-and-drop file attachment system connected directly to Project-IQ ITSM tickets.
- **Interactive Elements:**
  - Quick-reply buttons for guided flows
  - Star/Thumbs-up rating collection post-resolution
- **Session Persistence:** Remembers user conversation history across page reloads.
- **Input Guard Rails (Zero-Latency Security):**
  - Blocks PII (SSN, Credit Cards)
  - Filters profanity completely on the client side
  - Prevents spam/shouting (CAPS lock detection) and limits message lengths.

🛠️ 4. Advanced Admin Dashboard Capabilities

The Admin Dashboard provides total control over the bot's behavior, aesthetics, and performance metrics.

📊 Service Analytics & Reporting

- Visualizes total conversations, messages sent, and active users.
- Live charts tracking the most common user intents (e.g., Password Reset vs. Hardware Issues).
- Tracks user satisfaction ratings over time.

🔎 Conversation Inspector

- Real-time debugging of ongoing or past conversations.
- Advanced filtering: Search by User Name, App/Department (IT, HR), Ticket Status, and Time Period.
- Shows raw NLU confidence scores and extracted slots for every message.

🎨 Visual Theme Builder

- Change the look and feel of the chat widget live (Primary color, Header color, user/bot bubble colors).
- Instant live preview window.
- Themes are saved to PostgreSQL and instantly pushed to the embedded widget.

💻 AI Studio (In-Browser IDE)

- A fully integrated code editor (Monaco Editor) inside the browser.
- Edit AI training files (`nlu.yml` , `domain.yml` , `stories.yml`) safely.
- **Includes File Versioning:** Tracks backup history and allows one-click restoration of older files.
- **One-Click Retraining:** Trigger model training directly from the web interface without touching the server terminal.

📡 System Observability

- Real-time server health monitoring via `psutil` .
- Live tracking of CPU usage, Memory consumption, and Disk space.

🧠 5. Advanced AI Knowledge Harvesting

The chatbot doesn't just answer questions; it actively learns and builds the IT Knowledge Base through automated harvesting from both internal and external sources.

- **Automated Web Scraping via Firecrawl.dev:** Uses Firecrawl to turn entire vendor documentation websites (e.g., Microsoft, Cisco) into clean, LLM-ready markdown. This allows the bot to instantly ingest thousands of external pages into its internal semantic search database.

- **Unanswered Query Tracking:** Safely logs and flags user questions that the bot was unable to answer or find a matching internal KB article for.
- **Content Gap Identification:** Aggregates failed searches in the Admin dashboard, highlighting exactly what documentation the IT team needs to write next—or what sites to scrape via Firecrawl.
- **Resolution Mining:** Analyzes successful IT agent interventions (when a ticket is resolved manually) to suggest new automated flows and responses.
- **Dynamic Training Data Generation:** Allows admins to instantly convert real user utterances (from the Inspector) directly into new NLU training data (Intents & Stories) with a single click in AI Studio.

🔗 6. Project-IQ ITSM Integration (Full Ticket Lifecycle)

The system works seamlessly with Project-IQ ITSM to handle the entire lifecycle of an IT ticket.

- **Smart Search:** Before creating a ticket, the bot automatically queries the Knowledge Base using hybrid semantic search to attempt self-service resolution.
- **Ticket Creation:** Maps intents directly to dynamic ITIL Categories. Creates the formal ticket via REST API.
- **Status Updates:** Users can ask the bot "*What is the status of my ticket?*" and it pulls live database status from Project-IQ ITSM.
- **Updating Tickets:** Allows appending new notes or file attachments to an existing open ticket directly through chat.

🔒 7. Security & Infrastructure

- **API Authentication:** All Admin API routes are protected.
- **Safe Credentials:** Uses Infisical Vault so no tokens or database passwords are hardcoded in the repository.
- **Network Isolation:** The Flask API and Node static servers use specific port isolating, proxied securely through Nginx in production.
- **Robust Error Handling:** Fallback generic responses when external APIs (like Project-IQ ITSM) time out.

🎯 Summary

Project IQ is a **full-stack, production-ready AI solution**. It dramatically reduces Service Desk volume by capturing common issues, filtering malicious input, integrating deeply with Project-IQ ITSM, and providing IT Admins with a beautiful, real-time control panel to constantly improve the AI's intelligence.